

Simran Bilkhu

+1-289-400-6136 | bilkhs2@mcmaster.ca

[in my-profile](#) | [simb2](#) | [my-site](#)

Oakville, Ontario, Canada

EDUCATION

• McMaster University

Master's in Statistics

September 2024 - April 2026

Hamilton, Canada

◦ Grade: 11.86/12

◦ Thesis [\[link\]](#)

A Comparative Study of Bayesian Factor Models under PLT and UGLT Identification Constraints

• University of Toronto

Specialist in Applied Statistics with a minor in Math (Honours Bachelor's of Science)

April 2024

Toronto, Canada

◦ Cumulative GPA (final 2 yrs - 10.5 out of 20.5 total credits): 3.76/4.00

CURRENT RESEARCH

• Sparse Bayesian Dynamic Factor Models for Longitudinal Data (Thesis)

Ongoing

• Counterexamples to Continuity of Measure in Uncountable Unions

[Link to Preprint](#)

Submitted for review

GRADUATE PROJECTS & PRESENTATIONS

• Probability and Stochastics

Presentation

July 2025

[Link to Project](#)

◦ Prepared lectures on martingales, and their relationships to Markov processes, Weiner processes, and Poisson processes.

◦ Provided proofs of several theorems and properties.

◦ Included an introduction to stochastic kernels, and their connection to stochastic matrices for discrete state spaces.

• Bayesian Nonparametrics

Assignment

January 2025 - April 2025

[Link to Project](#)

◦ Provided a thorough proof showing that the proportions from the Chinese Restaurant Process converge in distribution to the GEM distribution.

• Bayesian Nonparametrics

Assignment

January 2025 - April 2025

[Link to Project](#)

◦ Showed that the size-biased permutations of the Poisson-Dirichlet distribution is given by the 2 parameter GEM distribution.

• Statistical Learning

Impacts of Diet Intervention on the Gut Microbiome

January 2025 - April 2025

[Link to Project](#)

◦ Applied dimension reduction via Principal Component Analysis (PCA) on microbiome count data with over 1400 features, and identified relevant taxa.

◦ Used model-based clustering techniques to explore group differences in microbial composition between plant and animal based diets to identify differing microbiome profiles.

◦ Identified important microbial taxa for dietary classification by analyzing loadings from principal components.

• Research Group Speaker

A Review of Sparse Bayesian Factor Analysis when the Number of Factors is Unknown

April 2025

[Link to Project](#)

◦ Reviewed and presented work by [Frühwirth-Schnatter et al.](#), highlighting a Bayesian approach to factor analysis with an unknown number of latent factors using spike-and-slab priors.

◦ Provided a background on shrinkage priors, UGLT structures, and [reversible jump MCMC](#).

- Research Group Speaker** January 2025
A Review of Sparse Principal Component Analysis [Link to Project](#)
 - Presented key theoretical links between factor analysis and PCA, highlighting the benefits of sparse representations for interpretability in high-dimensional settings.
 - Compared sparse PCA via penalized optimization (see [Zou et al.](#)) to varimax rotation, using simulation studies to assess interpretability and mean squared error across varying sparsity levels.

- Data Science** November 2024
Establishing Baseline Climate Partitions via Clustering [Link to Project](#)
 - Analyzed time-series raster climate data (temperature, precipitation, elevation) across Canada using python, focusing on spatial patterns and long-term climatic trends.
 - Performed K-means and Agglomerative clustering methods to group geographic regions using silhouette analysis to choose the number of clusters.
 - Visualized and interpreted clustering results through geographic heatmaps, and cluster-wise summaries.

- Data Science** September 2024
Expansion of Canadian Agriculture and the Relationship with Wetlands in Boreal Shield Provinces [Link to Project](#)
 - Conducted geospatial analysis of 66 GeoTIFF satellite land cover maps (2013–2023) using R's terra package to quantify agricultural land use and wetland area.
 - Preprocessed raster data into a tidy format and scaled estimates to square kilometers for year-by-year provincial comparison.
 - Designed and deployed a Shiny web app for interactive visualization, allowing users to explore province-specific trends and relationships across time ([link to site](#)).
 - Addressed limitations in remote sensing data such as non-random missingness due to satellite imaging challenges, and discussed implications for ecological monitoring and policy.

- Research Group Speaker** September 2024
Hamiltonian Monte Carlo [Link to Project](#)
 - Presented a conceptual and mathematical overview of Hamiltonian Monte Carlo (HMC), explaining the role of Hamiltonian dynamics.
 - Co-developed a Shiny web app allowing users to simulate from normal and beta distributions using HMC

WORK EXPERIENCE

- **McMaster University**, Statistical Methods and Applications January 2025 - April 2025
- **McMaster University**, Linear Regression Models Using SAS September 2025 - December 2025
- **Sodexo Canada**, IT Intern Mar 2020 - Aug 2022

WORKS CITED

- Frühwirth-Schnatter, Sylvia, et al. "Sparse Bayesian Factor Analysis when the number of factors is unknown (with Discussion)". *Bayesian Analysis*, vol. 20, no. 1, 2025, pp. 213–344.
- Zou, Hui, et al. "Sparse principal component analysis". *Journal of computational and graphical statistics*, vol. 15, no. 2, 2006, pp. 265–86.